

2303A52422

B-37

Assignment: 8.2

Topic: Generating and Working with Test Cases using Pytest

Task 1: Test-Driven Development for Even/Odd Number Validator

Task Description

The objective is to implement a function `is_even(n)` using a Test-Driven Development (TDD) approach with the pytest framework. We must identify edge cases (zero, negative numbers) and verify them using assertions.

Prompt

"I need to practice TDD using pytest. Generate a Python function `is_even(n)` and a corresponding set of pytest test functions. Ensure you test positive integers, negative integers, zero, and invalid inputs."

Code

```

def is_even(n):
    """
    Returns True if n is even, False otherwise.
    Validates that input is an integer.
    """
    if not isinstance(n, int):
        return "Invalid Input"
    return n % 2 == 0

# --- Pytest Test Cases ---
def test_positive_numbers():
    assert is_even(2) is True
    assert is_even(9) is False

def test_edge_cases():
    assert is_even(0) is True    # Zero is even
    assert is_even(-4) is True   # Negative even number

def test_invalid_input():
    assert is_even("string") == "Invalid Input"
    assert is_even(3.5) == "Invalid Input"

```

Output

```

===== test session starts =====
platform win32 -- Python 3.14.2, pytest-9.0.2, pluggy-1.6.0 -- C:\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\vidya\OneDrive\Desktop\ai coding
collected 3 items

ass8.py::test_positive_numbers PASSED [ 33%]
ass8.py::test_edge_cases PASSED [ 66%]
ass8.py::test_invalid_input PASSED [100%]

===== 3 passed in 0.06s =====

```

Explanation

Using pytest makes TDD cleaner. Instead of manually printing "Input | Result", we define functions starting with `test_`. Inside these functions, we use the `assert` statement. If `is_even(2)` returns `True`, the test passes silently. If it returns `False`, pytest would shout with a detailed error message. I separated tests into logical groups (positive, edge cases, invalid) to make debugging easier.

Task 2: Test-Driven Development for String Case Converter

Task Description

This task requires creating two functions, `to_uppercase` and `to_lowercase`. We will use pytest to handle "dirty" data—such as `None` values, empty strings, or numbers—ensuring the program doesn't crash.

Prompt

"Generate pytest test cases for string conversion functions `to_uppercase` and `to_lowercase`. Use assertions to verify they handle `None`, numbers, and empty strings safely without crashing."

Code

```

def to_uppercase(text):
    if text is None or not isinstance(text, str):
        return "Invalid Input"
    return text.upper()

def to_lowercase(text):
    if text is None or not isinstance(text, str):
        return "Invalid Input"
    return text.lower()

# --- Pytest Test Cases ---
def test_string_conversion():
    assert to_uppercase("ai coding") == "AI CODING"
    assert to_lowercase("TEST") == "test"

def test_empty_strings():
    assert to_uppercase("") == ""
    assert to_lowercase("") == ""

def test_invalid_types():
    assert to_lowercase(None) == "Invalid Input"
    assert to_uppercase(123) == "Invalid Input"

```

Output

```

cachedir: .pytest_cache
rootdir: C:\Users\vidya\OneDrive\Desktop\ai coding
collected 3 items

ass8.py::test_string_conversion PASSED [ 33%]
ass8.py::test_empty_strings PASSED [ 66%]
ass8.py::test_invalid_types PASSED [100%]

===== 3 passed in 0.02s =====

```

Explanation

Here, the tests act as a "contract" for the code. The test `test_invalid_types` specifically asserts that passing `None` or `123` must return "Invalid Input" rather than raising an exception. This enforces robustness. If I had written the function to just return `text.upper()` without checking the type first, `pytest` would have failed this test immediately, signaling that my code was unsafe.

Task 3: Test-Driven Development for List Sum Calculator

Task Description

We need to write a function `sum_list(numbers)` that calculates the total of a list. We will use `pytest` to verify that the function safely ignores non-numeric values.

Prompt

"Write a Python function `sum_list` and use `pytest` to verify it. The tests should check that strings inside the list are ignored and that empty lists return 0."

Code

```

def sum_list(numbers):
    total = 0
    for item in numbers:
        # Check if item is an integer or float
        if isinstance(item, (int, float)):
            total += item
    return total

# --- Pytest Test Cases ---
def test_standard_integers():
    assert sum_list([1, 2, 3]) == 6

def test_empty_list():
    assert sum_list([]) == 0

def test_mixed_types():
    # Should ignore "a" and sum 2 + 3
    assert sum_list([2, "a", 3]) == 5

def test_negative_numbers():
    assert sum_list([-1, 5, -4]) == 0

```

Output

```

cachedir: .pytest_cache
rootdir: C:\Users\vidya\OneDrive\Desktop\ai coding
collected 4 items

ass8.py::test_standard_integers PASSED [ 25%]
ass8.py::test_empty_list PASSED [ 50%]
ass8.py::test_mixed_types PASSED [ 75%]
ass8.py::test_negative_numbers PASSED [100%]

===== 4 passed in 0.08s =====

```

Explanation

The logic for `sum_list` is verified by four distinct tests. `test_mixed_types` is the most interesting one; it asserts that `[2, "a", 3]` equals 5. This confirms that our logic successfully filters out the string "a". If we used the built-in `sum()` function, this specific test would fail with a `TypeError`, guiding us to implement the manual loop with `isinstance` checks.

Task 4: Test Cases for Student Result Class

Task Description

This task involves Object-Oriented TDD. We will use `pytest` **fixtures** to create a fresh `StudentResult` instance for each test, ensuring clean state.

Prompt

"Create a `StudentResult` class. Use `pytest` fixtures to instantiate the class and write tests for adding marks, calculating averages, and determining Pass/Fail results."

Code


```

class StudentResult:
    def __init__(self):
        self.marks = []

    def add_marks(self, mark):
        if mark < 0 or mark > 100:
            raise ValueError("Mark must be 0-100")
        self.marks.append(mark)

    def calculate_average(self):
        if not self.marks:
            return 0
        return sum(self.marks) / len(self.marks)

    def get_result(self):
        avg = self.calculate_average()
        return "Pass" if avg >= 40 else "Fail"

```

```

# --- Pytest Fixtures & Tests ---
@pytest.fixture
def student():
    return StudentResult()

def test_pass_scenario(student):
    for m in [60, 70, 80]:
        student.add_marks(m)
    assert student.calculate_average() == 70.0
    assert student.get_result() == "Pass"

def test_fail_scenario(student):
    for m in [30, 35, 40]:
        student.add_marks(m)
    assert student.calculate_average() == 35.0
    assert student.get_result() == "Fail"

def test_invalid_mark_raises_error(student):
    # Verify that invalid input raises an exception
    with pytest.raises(ValueError):
        student.add_marks(-10)

```


Output

```
collected 3 items

assign-8.py ...

===== 3 passed in 0.02s =====
```

Explanation

In OOP testing, state is everything. If we reused the same student object for all tests, marks from the first test might mess up the second test. I used the `@pytest.fixture` decorator to create a fresh student object automatically for every test function. I also used `pytest.raises(ValueError)`, which is a professional way to test that code *correctly* errors out when given bad input (like -10).

Task 5: Test-Driven Development for Username Validator

Task Description

The final task uses `pytest.mark.parametrize` to run the same test logic on multiple different inputs (valid and invalid usernames) efficiently.

Prompt

"Write a username validator function. Use `pytest.mark.parametrize` to test valid inputs ('user01') against invalid ones ('ai', 'user name', 'user@123')."

Code

```
import pytest

def is_valid_username(username):
    # Rule 1: Length >= 5
    if len(username) < 5:
        return False
    # Rule 2: Alphanumeric only (no spaces, no symbols)
    if not username.isalnum():
        return False
    return True

# --- Pytest Parametrized Testing ---
@pytest.mark.parametrize("username, expected_result", [
    ("user01", True),      # Valid
    ("ai", False),         # Too short
    ("user name", False),  # Contains space
    ("user@123", False),   # Contains symbol
    ("admin123", True)     # Valid
])
def test_username_validation(username, expected_result):
    assert is_valid_username(username) == expected_result
```

Output

```
cachedir: .pytest_cache
rootdir: C:\Users\vidya\OneDrive\Desktop\ai coding
collected 5 items

ass8.py::test_username_validation[user01-True] PASSED [ 20%]
ass8.py::test_username_validation[ai-False] PASSED [ 40%]
ass8.py::test_username_validation[user name-False] PASSED [ 60%]
ass8.py::test_username_validation[user@123-False] PASSED [ 80%]
ass8.py::test_username_validation[admin123-True] PASSED [100%]

===== 5 passed in 0.09s =====
```

Explanation

Instead of writing five separate test functions, I used `@pytest.mark.parametrize`. This allows us to feed a list of inputs and expected outputs into a single test function. It makes the code much cleaner and easier to expand. If we wanted to test a new case, like "super_user", we would just add one line to the list, and pytest would automatically generate a new test case for it.